

**Time:** 6 minutes**Outcome:** Calculate private, public, and national saving in closed and open economies.**Difficulty:**

Intermediate

Saving, Investment, and National-Saving Identities



THE CORE IDEA

Let Y be income, T net taxes, C consumption, G government purchases, I investment, and S national saving. Private saving is $Y - T - C$; public saving is $T - G$. Thus $S = Y - C - G$. In a closed economy, $S = I$. In simplified open-economy accounting without net income or transfers from abroad, $S = I + NCO$ and $NX = NCO$, where NCO is net capital outflow and NX is net exports.



HOW TO RECOGNIZE IT

- Budget surplus means $T - G$ is positive; a deficit makes public saving negative.
- Adding private and public saving cancels T and gives $Y - C - G$.
- Use $S = I$ only for the closed-economy model.
- In an open economy, saving can finance domestic investment or net capital outflow.



WATCH OUT

Do not reverse the public-saving sign. $T - G$ is public saving; $G - T$ is the budget deficit. Also state the closed-economy assumption before writing $S = I$.



WORKED EXAMPLE: FROM COMPONENTS TO OPEN-ECONOMY SAVING

Saving and its allocation (dollars)

	Calculation	Value
Private saving	$1200 - 150 - 760$	290
Public saving	$150 - 260$	-110
National saving	$290 - 110$	180
Domestic investment	Given	140
NCO and NX	$180 - 140$	40

Let $Y = \$1,200$, $T = \$150$, $C = \$760$, and $G = \$260$. Private saving is \$290; public saving is -\$110; national saving is \$180. If domestic investment is \$140, then $NCO = S - I = \$40$. The accounting identity also gives $NX = \$40$.



CHECK YOURSELF

Private saving is \$220 and public saving is -\$50. What is national saving?

**READY?**

Return to the game and master this concept.